Green Chemistry

- 12 principles of green chemistry
- ✱ atom economy [application, calculation, example]
- Green synthesis [examples]
- Green solvents and green reagents
- Coining of green chemistry
- Nitrogen cycle
- soil & microbes

Water chemistry

- Water pollution
- hardness of water [numericals]
- methods of removal of hardness of water [Zeolite] (Chemmissible) [ampi or tona]
- BOD & COD [how to measure] [examples]
- Scale & sludge and other simple terms.
- Unique properties of water
- Heavy metal toxicity [examples] - Minimata
- Application of Nano science
- waste water management

Unit III

- IR, UV, NMR [principle, block diagrams]
- TGA, DTA, DSC
- R_f values for chromatography
- Table for IR [frequency] - wave number table
- NMR - principle, block diagram (numerical)
- UV - principle - hipsochromic shift & Bathochromic
- GCMC - block diagram, Mass spectra
- Beer Lambert's law
- finger print 400-700
- 700-1200 - normal stretching
- 1000-13000 - hydrogen stretching

L C-H stretching ⇒ 2900 - 3100
↳ N-H peak ⇒ 3400
↳ O-H peak ⇒ 3400
↳ thermal methods
↳ TGA, DTA, DSC - principle and application

Pollution
↳ case studies - Bhopal gas, fukushima power plant, chernobyl
↳ Solid waste management
↳ CFC
↳ eutrophication

⇒ VSEPR Theory
⇒ Transport number ⇒ important
⇒ hittorf's method

ronmental Chemistry & Pollution

↳ Air, water, soil, marine, Noise, Thermal, Nuclear pollution

↳ Solid waste management - causes, effects, control measures of urban and industrial waste

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Pollution - Addition of undesirable material into the envir. due to human activities.

⇒ pollutants - agents which cause pollution are called pollutants. They are physical, chemical or biological agents

Types of Pollution

1) Air pollution - pollutants when enter the air, cause air pollution. Due to the industries, vehicles etc, harmful gases are released in the air.

Air pollutants

1) suspended Particulate matter
2) gases like CO_2, NO_2
3) Fly ash

(i) Particulate pollutants : size ranges from 0.001 to 500 um.
Major sources : vehicles, industries.

(ii) Fly ash : Ejected by thermal power plants due to the burning of coal. Causes heavy metal pollution in water.

(iii) Lead and other metals : Lead mixed with water & food can cause poisoning. effects kidney and liver

(iv) Gaseous Pollutants - These include gases like CO_2, SO_2 etc which effects the humans and plants in diff ways like causing acid rain, green house effect, resp. diseases

Indoor Pollution

Causes

1) Poor ventilation
2) Usage of disinfectants and fumigants
3) burning of firewood (mainly in rural areas)
4) Compounds released from paints, furniture etc.

Preventive measures

1) opening up windows for ventilation
2) Use of biogas.
3) proper waste disposal
4) keeping home well sanitized.

Industrial pollution

1) Installing devices
   1) Filters - remove particulate matter
   2) ESP - remove ionized particulate matter.
   3) Scrubbers - They remove aerosols from a stream of gas
2) Use of cleaner fuels like CNG & LNG.
3) Environment friendly processes.
4) Control of vehicular pollution

Ozone layer depletion

When Chlorine is released from chemicals such as CFCs, this chlorine reacts with O_3 molecules present in the Ozone layer and forms O_2 molecule and nascent. Thus the amount of ozone gets reduced and entry of UV rays cannot be prevented.

Global warming and Green house

Gases like CO_2 and methane traps the heat radiated which causes the temperature of the earth to rise causing melting of polar ice caps.

2) Noise Pollution

Undesirable sounds or noise which can cause harm to the body is called noise pollution

Anything above 80dB of noise is hazardous

Sources of noise pollution :-

a) Indoor sources - noise produced by television, radio, refrigerator, transport etc.

b) Outdoor sources - use of loudspeakers at functions, vehicles, horns, industries, market place

effects of noise pollution

- disturbed sleeping patterns
- headaches
- insomnia
- depression
- hypertension

Prevention for noise pollution
- limited use of loudspeakers
- using silencer in vehicles and not using horn unne[illegible]
- using sound proof system in industries
- making properly welded rail tracks.

3) Water pollution : When pollutants enter the water bodies and cause undesirable effects is called water pollution.

Sources of water pollution
1) fertilizers & insecticides used in fields
2) release of chemical waste from industries
3) Human excreta and other waste released in water.
4) flyash or solid waste
5) Metals like zinc, mercury and cadmium.

Mercury causes - minimata disease
Lead causes - dysplexia disease
Cadmium causes - Itai Itai disease

Thermal pollution
The used hot water from industries is directly released into the water bodies which increases the temperature of the water which decreases the dissolved oxygen in water and harms marine animals.

preventive measures
1) Storing hot water in cooling ponds
2) Allowing water to cool before discharge.

Eutrophication - very important

When waste from fields, drainage, industries enter into water, they make the water enriched with nutrients. Due to this, it leads to growth of algae, phytoplanktons and other aquatic plants. These use up the dissolved oxygen present in the water and BOD increases. When they die, the DO in water becomes very less causing death of marine animals.

preventive measures

1) sedimentation, coagulation, filtration & disinfection are primary methods.
2) Water should be recycled and reused
3) quantity of waste water discharge should be minimized

4) Soil Pollution

When pollutants enter into the soil, they cause soil pollution

Sources of soil pollution

1) plastic bags
2) fertilizers, pesticides and insecticides
3) Industrial waste

preventive measures

1) minimising use of plastic bags
2) Sewage should be treated properly before using as fertilizer
3) Industrial waste should be properly treated

5) Radiation Pollution

Radiation produced from nuclear power plants and nuclear wastes cause radiation pollution.

It is of two types:

1) Non-ionizing
2) Ionizing

o There are two types of radiation damage

1) somatic damage [cells]
2) genetic damage [genes]

Case Study of Bhopal Gas Tragedy

In 1984, methyl isocyanate escaped from a plant which spread to the neighbourhood areas causing deaths of around 15,000 - 20,000 people and also killed plants and animals.

Case study of Chernobyl

In 1986, a large amount of radioactive material got released from a nuclear power plant in Chernobyl. People of the city had to flee and it caused deaths of workers and other people.

* Solid waste management

1) Sources - household, Industries, construction areas, Plants & sites
2) Effects - health hazards, unclean surroundings, soil pollution
3) Prevention - properly treated landfills, incineration [burning at higher Recycling, Composting

X —— X —— X

12 principles of green chemistry

1) Better to prevent waste rather than cleaning.
2) Synthetic methods to maximize incorp. of all materials [Atom economy]
3) Synthetic methods should be designed causing noor minimal toxicity.
4) Chemical products should be made for purpose while reducing toxicity
5) Use of Auxillary substance (solvents) should be stopped
6) Recognizing energy required and cost for the same
7) Renewable raw material should be used
8) stopping unnecessary derivitization.
9) Using catalytic reagents
10) Break down of final product should be easy
11) Developing Analytical methodologies
12) Minimising chemical accidents like releasing, explosions.

Atom Economy

↳ Developed by Barry Trost in 1991
↳ Method of expressing the efficiency in a rxn.

$$\boxed{\% \text{ yield} = \frac{\text{Actual yield of product}}{\text{theoretical yield of product}} \times 100}$$

↳ A rxn may have 100% yield but may not be considered as Green synthesis if it produces large number of by-products.

↳ $$\boxed{\% \text{ atom economy} = \frac{\text{Mass of atom in desired product}}{\text{mass of atoms in reactants}} \times 100}$$

=) Rearrangement rxn ~~??~~ - 100%
Addition rxn - 100%
Oxidation rxn

Green Solvents

↳ Environmentally friendly solvents
↳ derived from processing agricultural crops.
↳ Example => ethyl lactate [best]
↳ high solvation power & eco-friendly

Green Reagents

using reagents which donot pollute environment

What is Green Chemistry all about.

↳ Use of non-toxic substances
↳ Atom economy
↳ Minimizing waste product
↳ Using renewable resources
↳ Cutting cost and energy efficiency

Soil microbes and functions
↳ Soil Bacteria - Breaks down cellulose and chitin
↳ Soil actinomycetes - Breaks down organic compounds
↳ Soil fungi - Breaks down cellulose
↳ Soil protozoa - releases nitrogen that can be used by plant

chemistry

Hardness of water

Presence of salts of calcium and magnesium in water is called hard water. It does not lather up with soap and forms precipitate

Types

1) Permanent Hardness - Chloride and sulphate salts of Ca & Mg present in water constitute to permanent hardness

methods to remove

a) Permutit Method (Zeolite method)

$NaZ + Ca^{++} \rightarrow CaZ + Na^{+}$

$Z = AlSiO_4$

b) Calgon's Method

In this method we use $Na_6P_6O_{18}$ reagent known as calgon. Ca^{++} & Mg^{++} ~~and~~ ions get adsorbed and hardness is removed

2) Temporary Hardness - Carbonate and bi-carbonate salts of Ca & Mg in water constitute temporary hardness

Methods to remove

a) Boiling.

b) Clarke's method - Adding accurate amount of clarke's reagent i.e $Ca(OH)_2$

Degree of hardness

$$= \frac{[\text{Strength of substance producing hardness}]\,[\text{eq. weight of } CaCO_3]}{[\text{eq. weight of hardness producing}}$$

Boiler Trouble

Boilers are used to convert water into steam.

(a) Scales formation ⇒ hard deposits which stick on the inner walls of the boiler. eg ⇒ $CaCO_3$, $Ca(HCO_3)_2$. Can be removed by phosphate conditioning

(b) Sludge formation ⇒ soft deposits in the colder area of the boiler. eg - $CaCl_2$, $MgCl_2$

(c) caustic embrittlement - corrosion of boiler

(d) Priming and foaming

Chemical Oxygen Demand (COD)

The amount of oxygen that is required for chemical oxidation of organic & inorganic chemicals present in the waste water. It is determined by chemical oxidation of water with dichromate

Biological Oxygen Demand (BOD)

Amount of oxygen required by microorganisms in the sewage for decomposition of bio-degr. matter in presence of air. It is determined by incubating a water sample and measuring the decreasing dissolved oxygen as bacteria decompose the material.

iqu properties of water

↳ polar in nature [able to dissolve and dissociate many particles]
↳ water boils and freezes slowly. otherwise drastic changes would start taking place in nature.
↳ universal solvent
↳ has high heat capacity
↳ less dense as a solid as in a liquid.

Heavy metal toxicity

↳ Lead, Mercury, Cadmium etc
↳ May enter our body through food, water or air
↳ causes various diseases like Mini-mata, displexia and 9tai-9tai.
↳ preventive measure
  ↳ proper disposal of industrial waste into water
  ↳ using catalytic converters.

↳ Mercury - causes Mini-mata - toxic involatile form
Cadmium - Causes 9tai-9tai - highly toxic
~~dark~~ lead - causes displexia - severely toxic

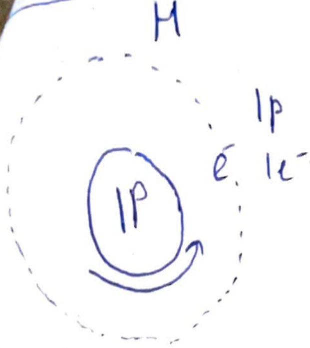
what is heavy metal

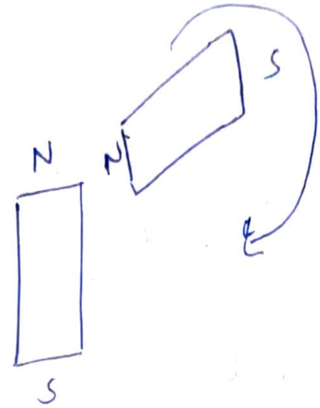
Metal whose density is more than 5g/cm3.

what causes heavy metal toxicity

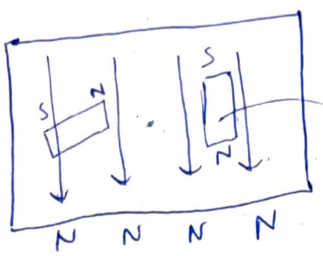
↳ Agriculture
↳ industries
↳ humus having high affinity towards metals

MR Spectroscopy





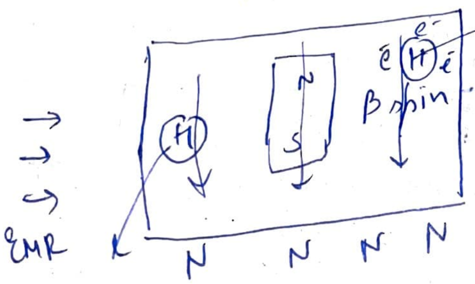
H- acts as a magnet



rearrange to achieve stable state

→ α - spin

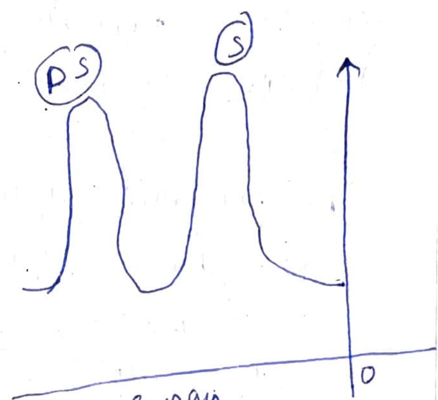
⇒ Gives idea about how other atoms are placed with respect to this position.



→ surrounded by e-
so effect of M.F is low
so less energy required for β-spin

effect of M.F is high, so more energy is required

C - H	C -- H - C
only one C, so less shielded	2-C, so more shielded
more energy required	less energy required
unshielded	shielded



Bases on the energy required to reach β-state, we judge whether it is shielded or deshielded.

IR Spectroscopy

helps us to find out the structure of different chemi[cal] groups - mostly the functional groups.

↳ IR radiations are used.

Source ⟶ Sample [Liq / Solid / Gas] ⟶ Detector

Source ↓ IR radiations ↓ different wavelength

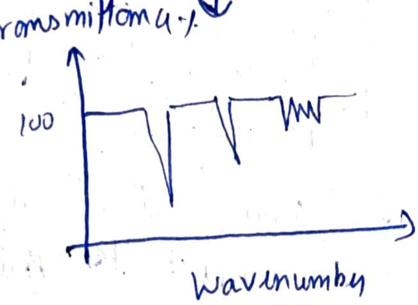
Sample ↓ Energy absorb

H-C(=O)-H

↓

Vibrate

Detector ↓ transmittance %



Vibrate → Stretching, Bending

Stretching → Symmetric stretching, Asymmetric stretching

Bending → wagging, rocking, scissoring, Twisting

Finger print region → 1600 - 400 cm^{-1}

Absorption region → 4000 - 1800 cm^{-1}

No. of modes of vibration

for linear ⇒ 3N-5

for Non-linear ⇒ 3N-6

	Wave Number
C-H	⇒ 3300 - 2700
N-H	⇒ 3500 - 330
O-H	⇒ 3650 - 3200
C-O	⇒ 1250 - 1050
C=O	1780 - 1650
C=C	⇒ 1680 - 1600
C≡C	⇒ 2260 - 2100

⇒ Differential scanning calorimetry

Sample

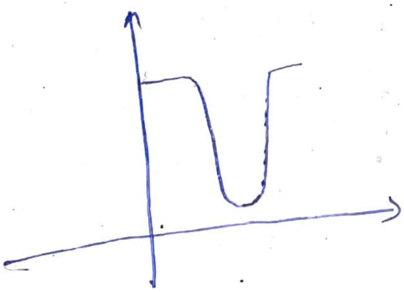
Reference

$\boxed{\text{We need to maintain the temperature in both the pans}}$

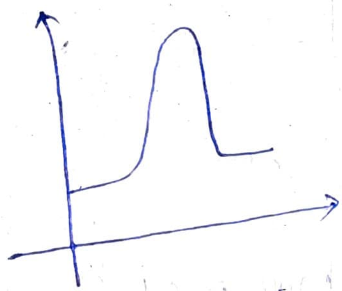
⇓ either endo or exothermic

↳ Technique for qualitative and quantive analysis of the analyte

↳ The amount of heat provided to maintain the temperature of the sample & reference gives us idea about quality & quantity



Exothermic

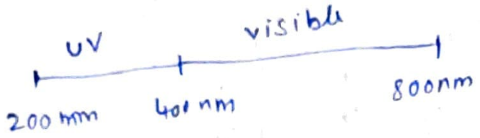


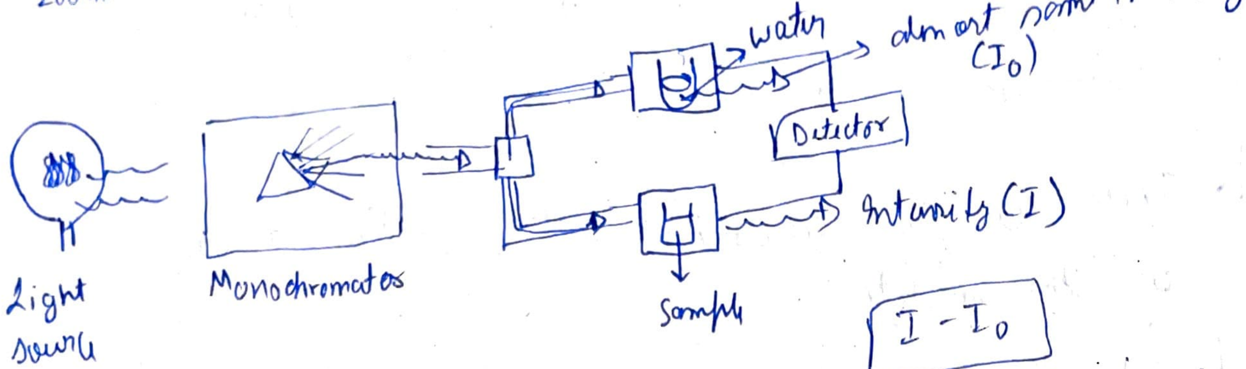
Endothermic

UV spectroscopy

↳ For concentration of different substances, molecules or ...

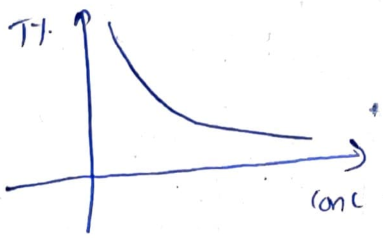
↳ utilizes UV or visible range of light.





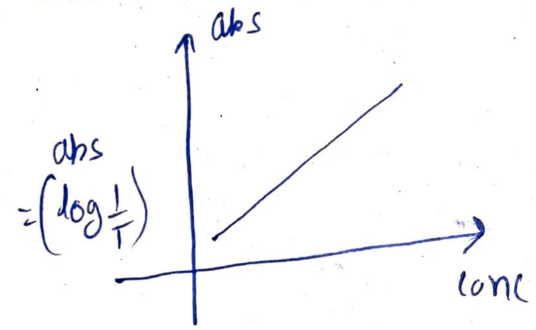
$\boxed{I - I_0}$

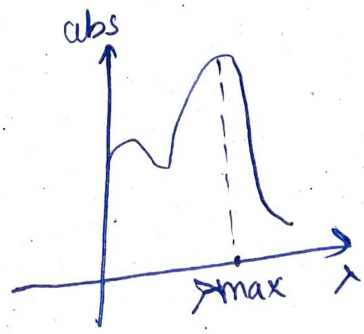
$\boxed{T = \frac{I}{I_0}}$



$T = 10^{-cl}$

⇒ $\boxed{\text{Absorbance} \propto cl}$ - Beer-Lambert law





Qualitative

Quantitative

$\boxed{\text{abs} = \varepsilon \times c \times l}$